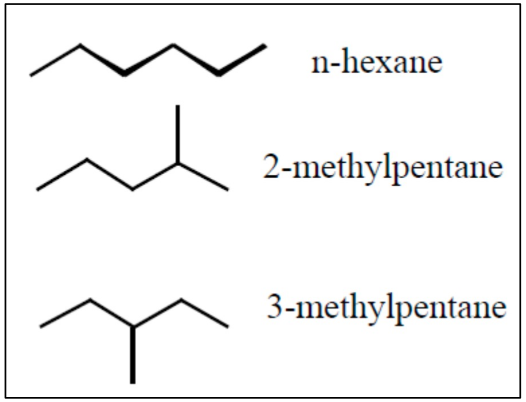
Stereochemistry and applications

What are isomers?

Structural isomers

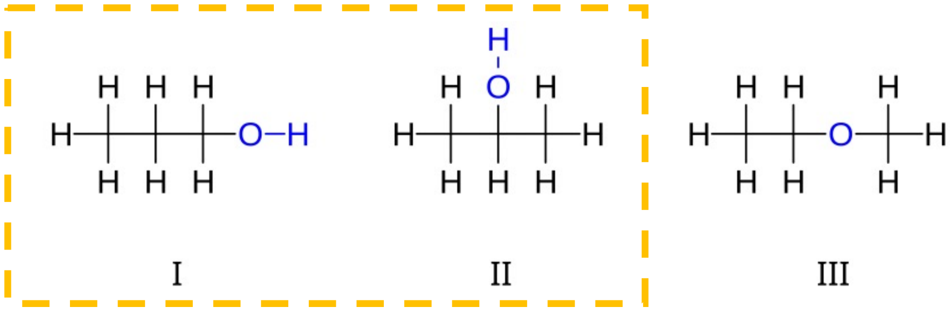


Identical molecular formula (same number of atoms of elements)

Hydrocarbon of the formula C_6H_{14}

Same type of bonds, different connectivity

Compound C_3H_8O

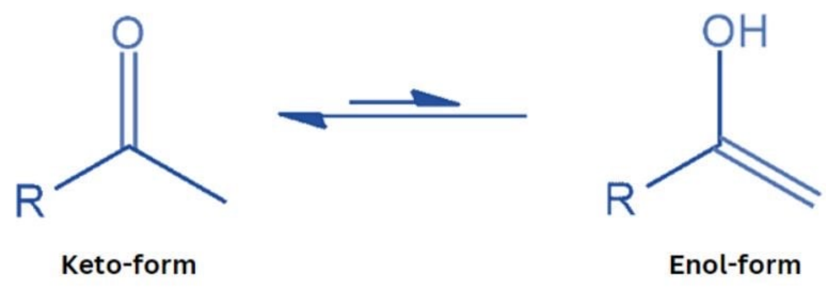


Regioisomers

Different type of bonds, different connectivity

Same functional group, different positions

Keto-Enol Tautomerism



Structural isomerism

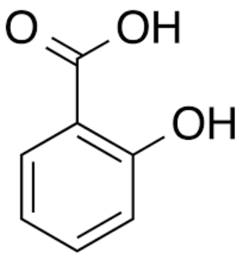
Isomers may differ in

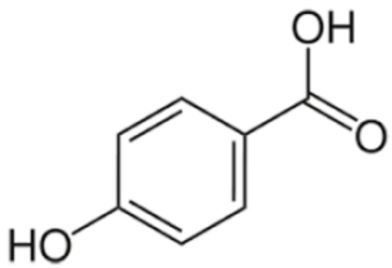
Chemical properties (nature of reactivity, rates of reactions, end product)
Physical properties (melting point, boiling point, optical properties)

Both chemical and physical properties

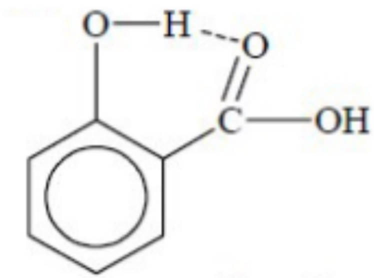
Regioisomers: P-hydroxy benzoic acid vs salicylic acid

P-hydroxybenzoic acid has higher boiling point than salicylic acid. Explain why?

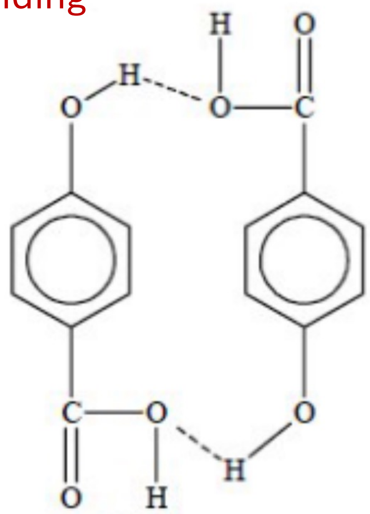




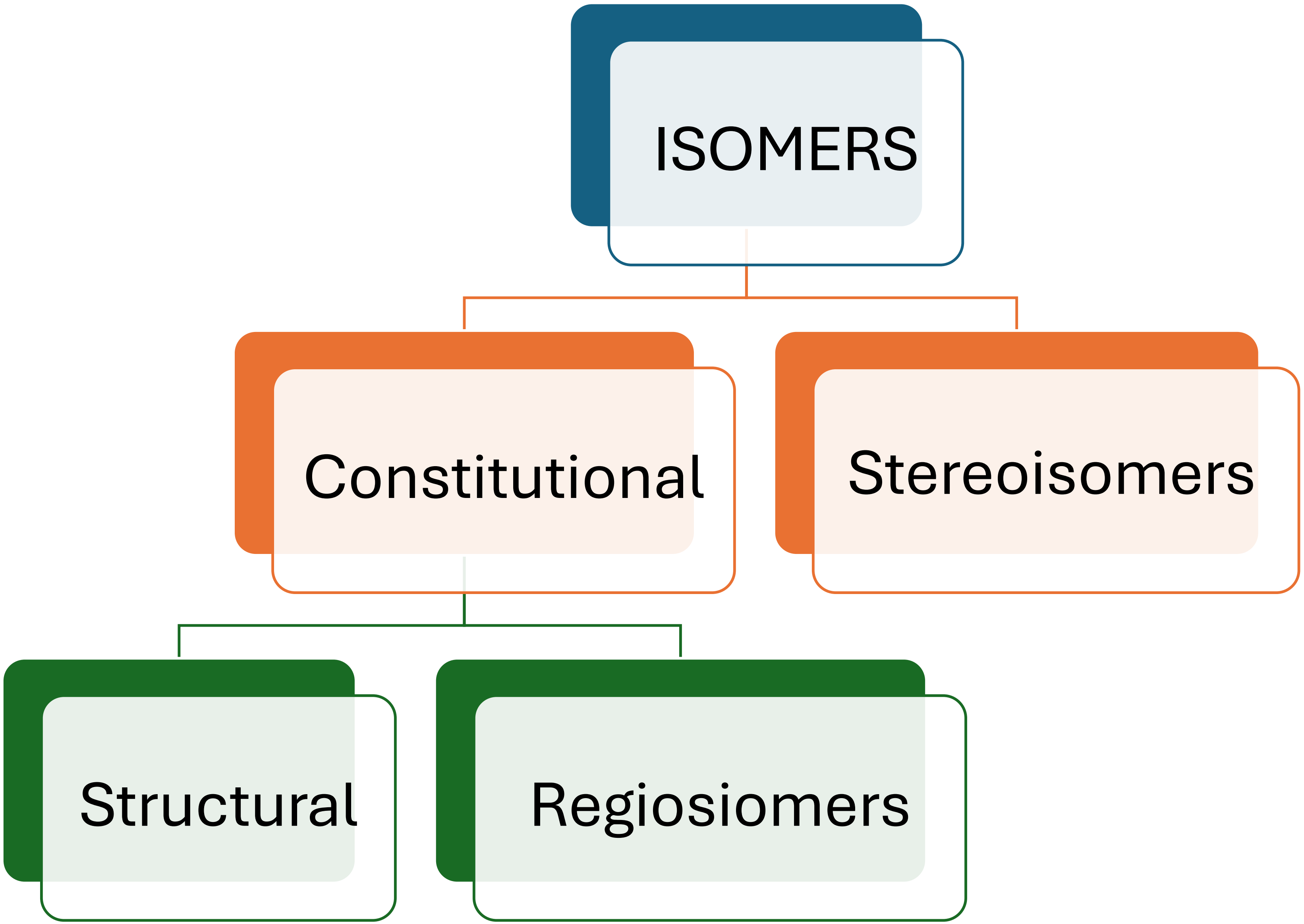
Inter vs Intra molecular H-bonding



intramolecular Hydrogen bonding in O- hydroxy benzoic acid

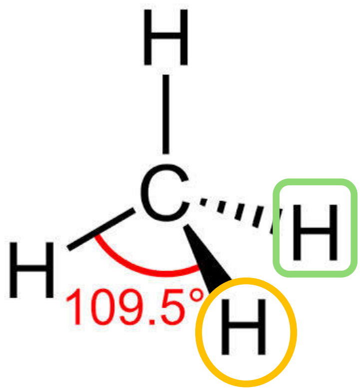


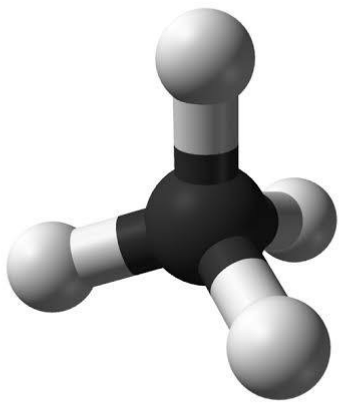
inter molecular Hydrogen Bonding in P-Hydroxy benzoic acid.


ISOMERS
Constitutional
Stereoisomers
Structural
Regiosiomers

What are stereoisomers?

Same connectivity but different arrangement of bonds in space

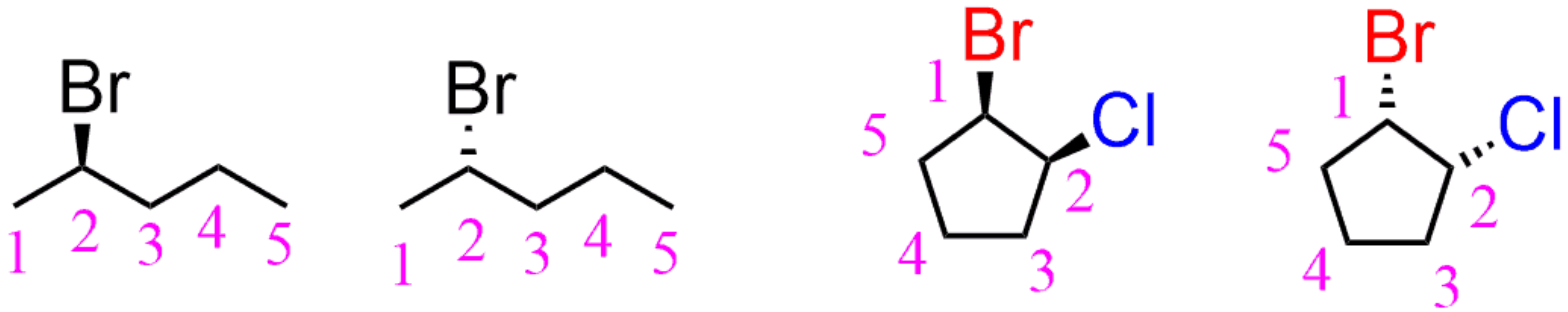




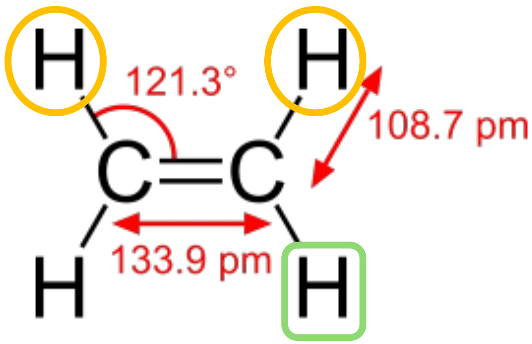
One hydrogen above the plane

Another hydrogen is behind the plane

Stereoisomers - same connectivity, different arrangement

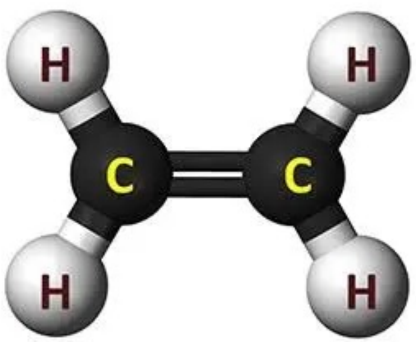


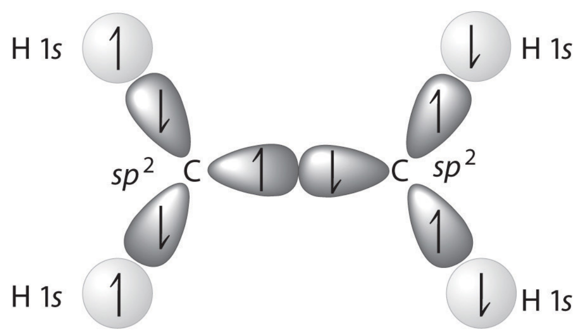
Ethylene



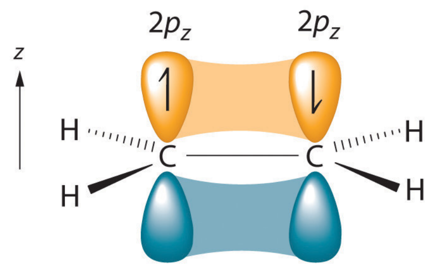
Yellow circled hydrogens are on the same side

Yellow and green hydrogens are on the opposite side

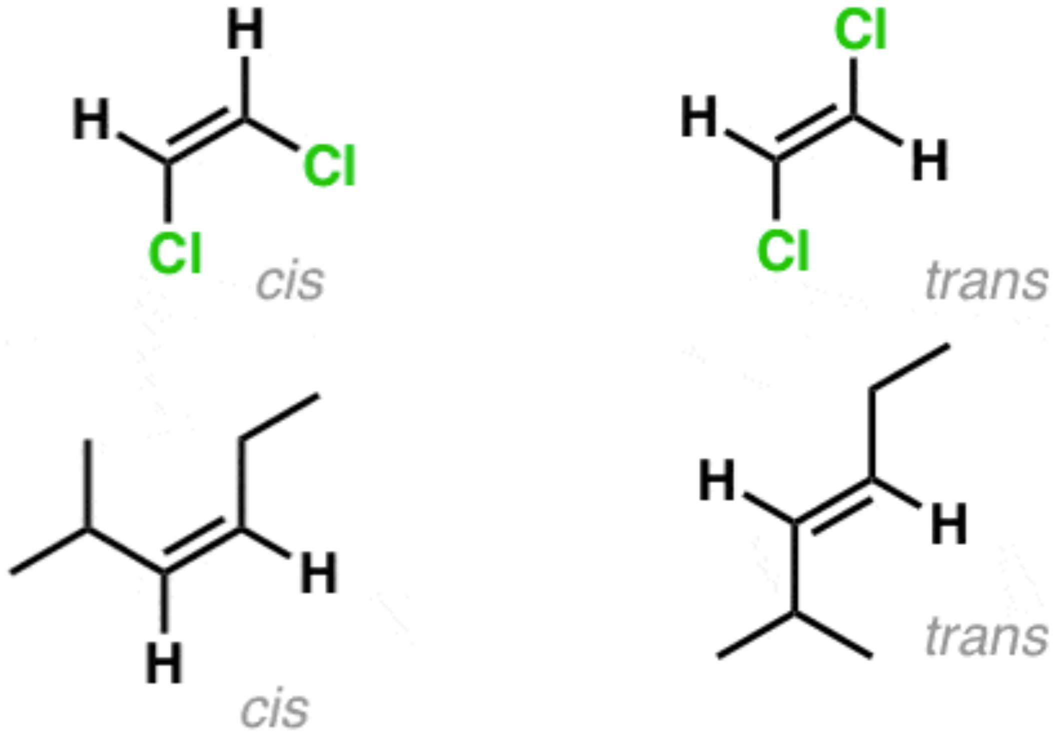




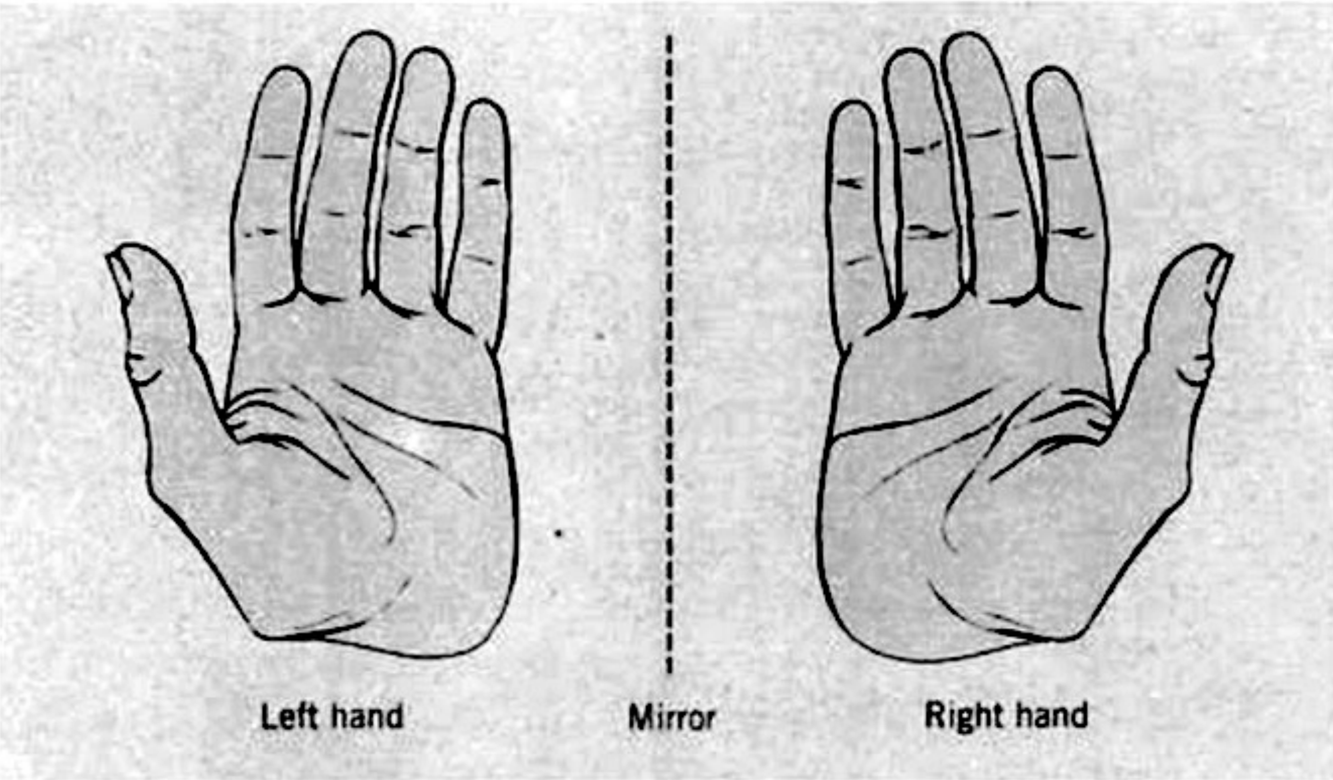
(a) C_2H_4 σ-bonded framework

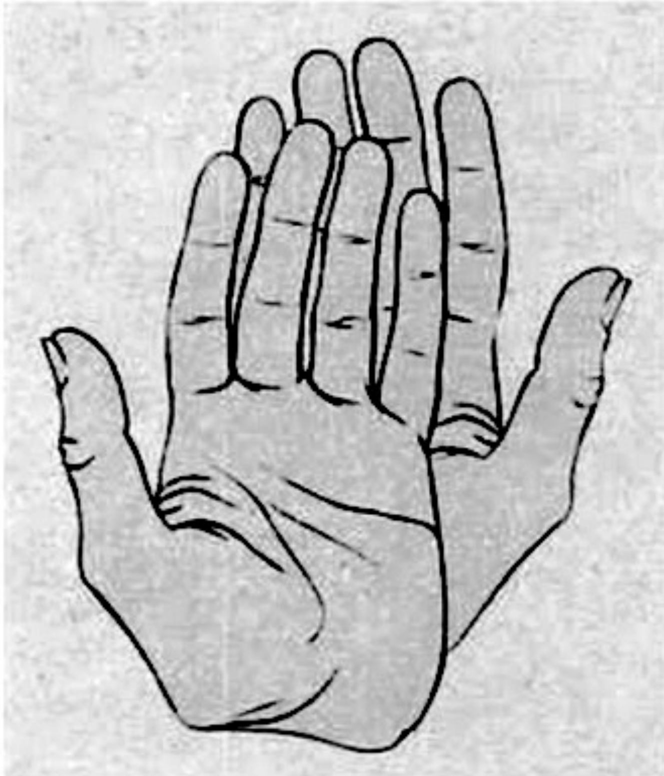


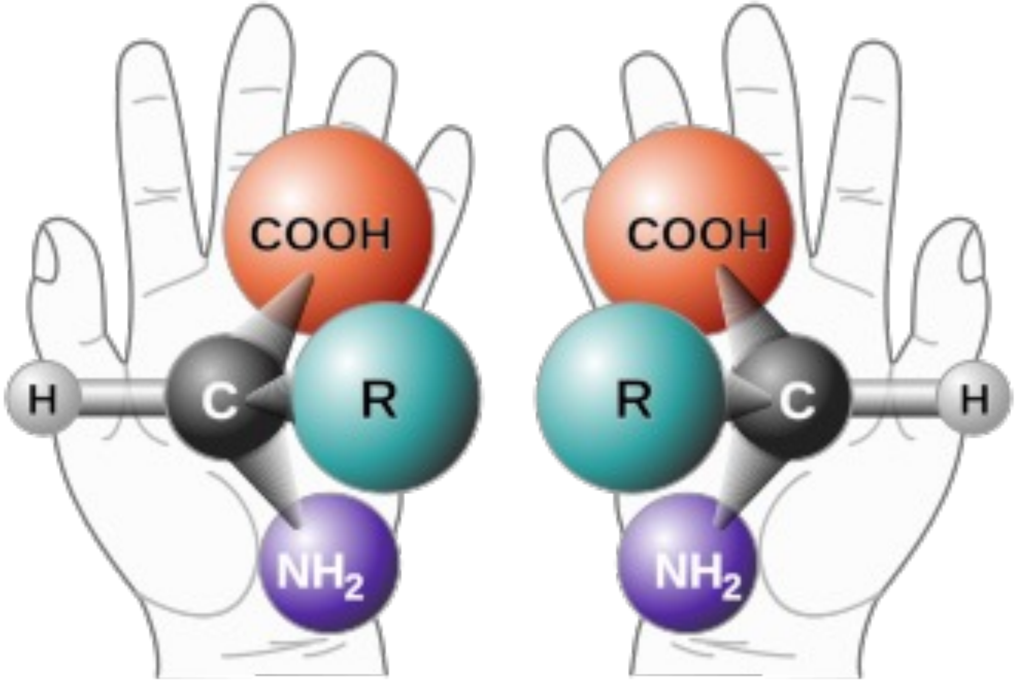
(b) C_2H_4 π bonding

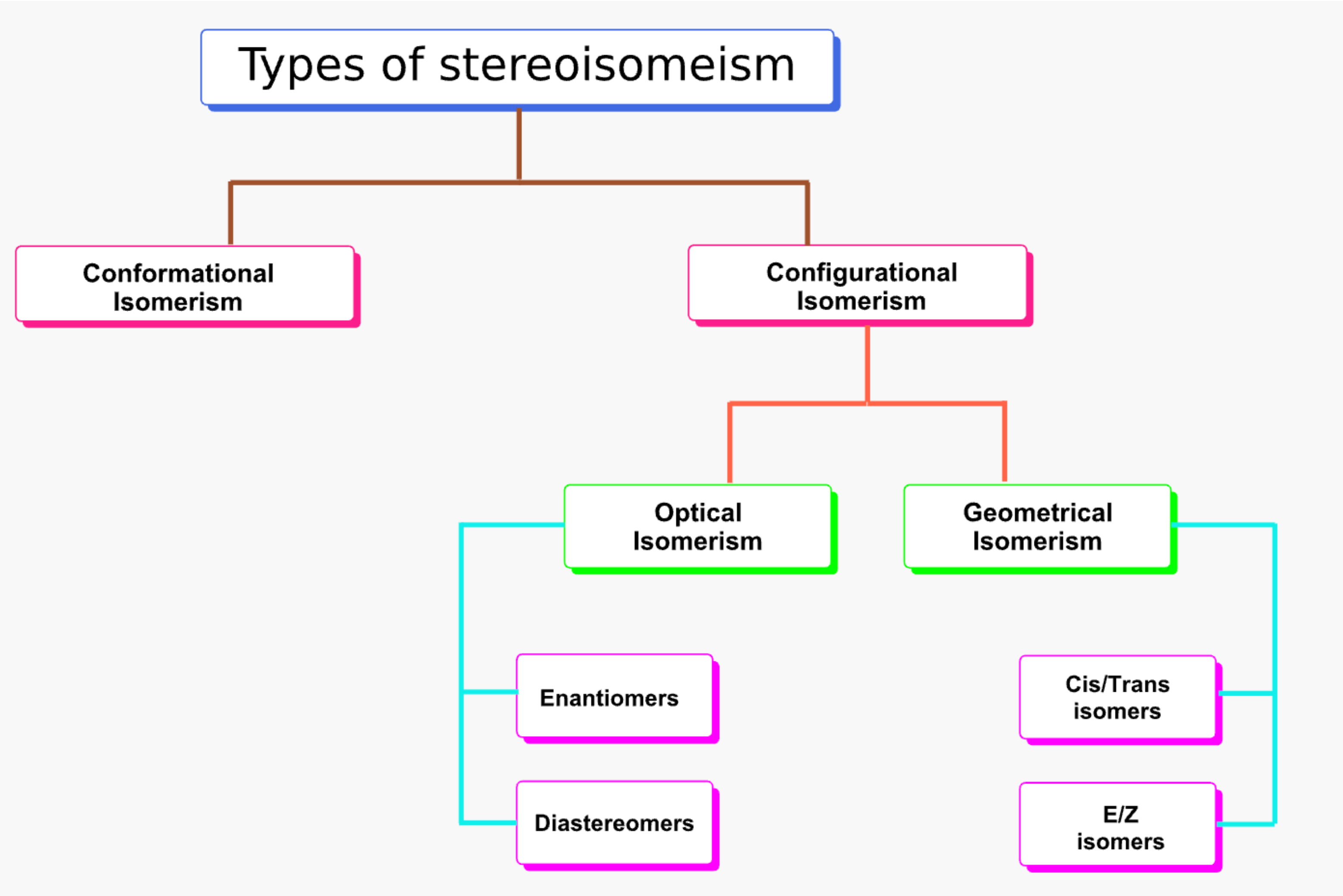


pi bonds cannot be rotated – no free rotation unlike single bonds


Left hand
Mirror
Right hand




COOH
H
C
R
NH2
COOH
R
C
H
NH2

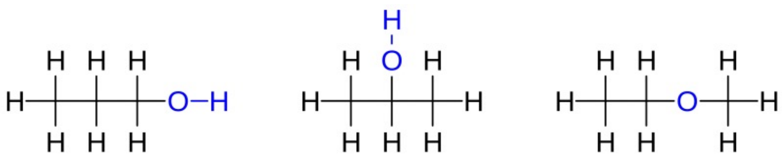

Types of stereoisomeism
Conformational Isomerism
Configurational Isomerism
Optical Isomerism
Geometrical Isomerism
Enantiomers
Diastereomers
Cis/Trans isomers
E/Z isomers

Constitutional isomers vs Stereoisomers

May have different bond connectivity

Always different properties
(Both physical and chemical)

If functional groups are different, then drastic change in properties



I — 1º alcohol

II — 2º alcohol

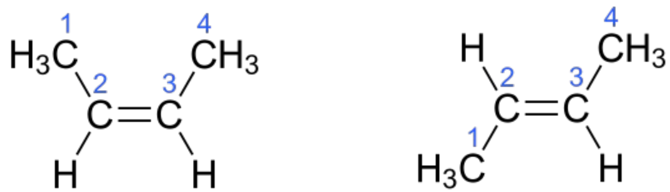
III — ether

Same bond connectivity

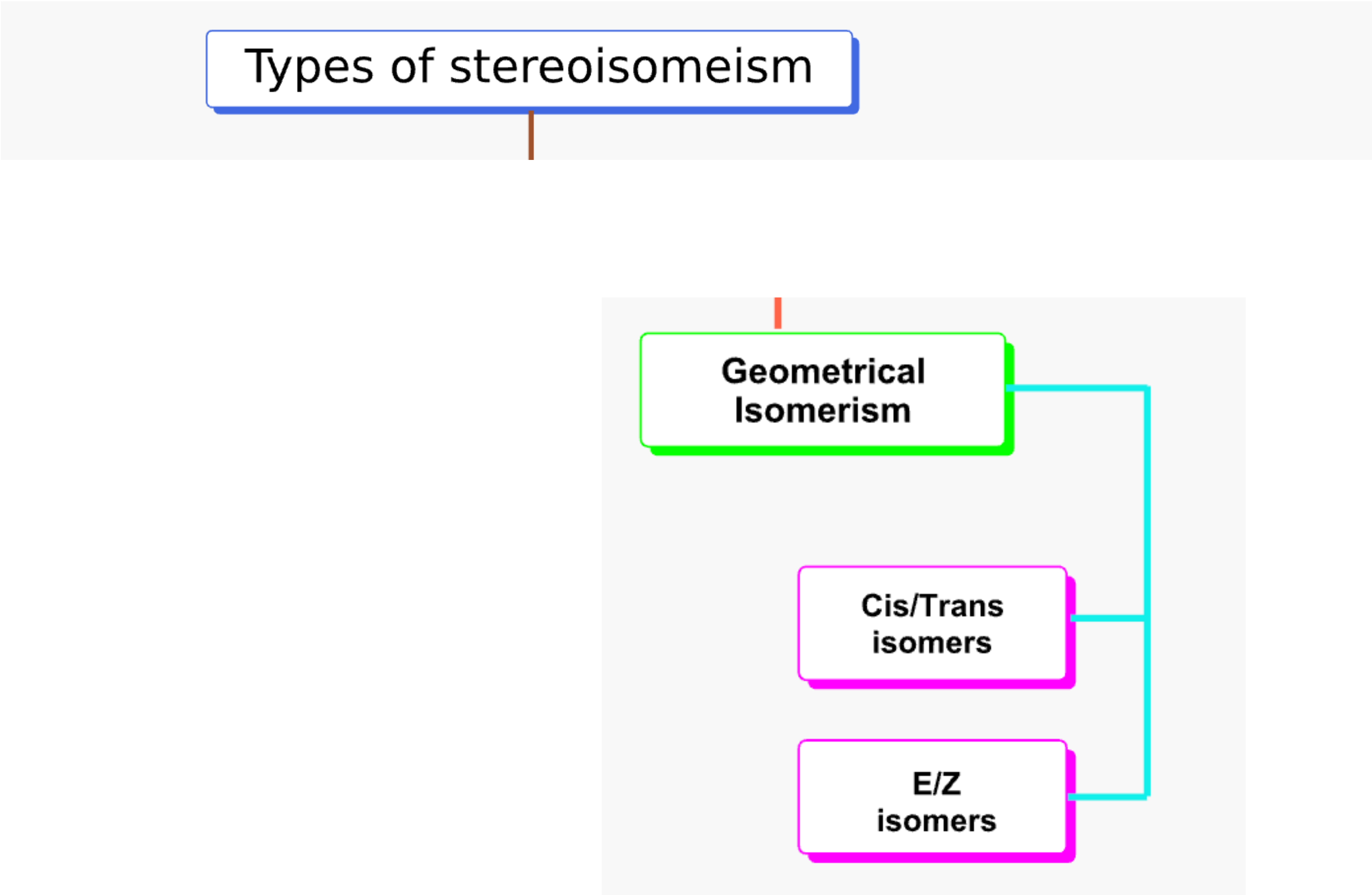
Different arrangement in space

May or may not have different chemical reactivities

May or may have different physical properties
(melting points, boiling points, solubility, **optical** etc.)

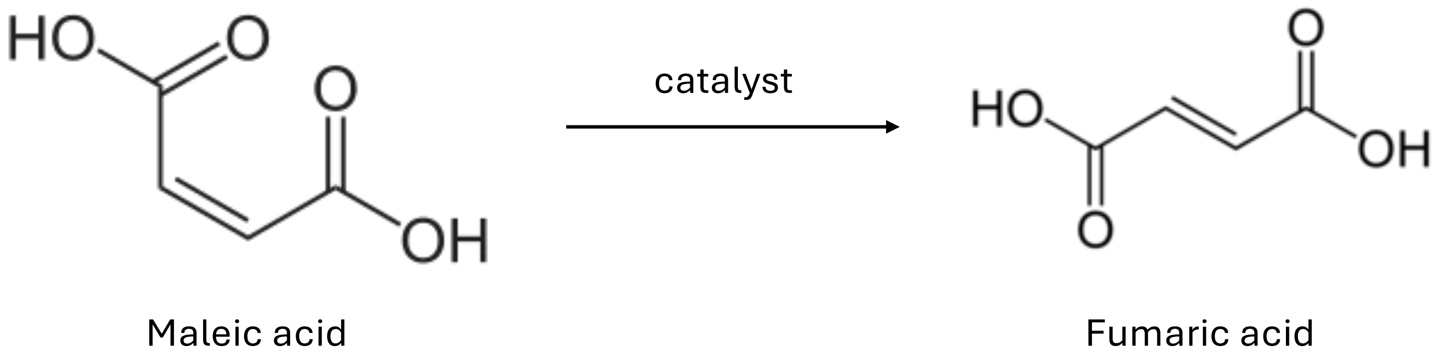


Trans isomers have higher melting point than cis isomers because the molecules are more symmetrical and are tightly packed in a lattice


Types of stereoisomeism
Geometrical Isomerism
Cis/Trans isomers
E/Z isomers

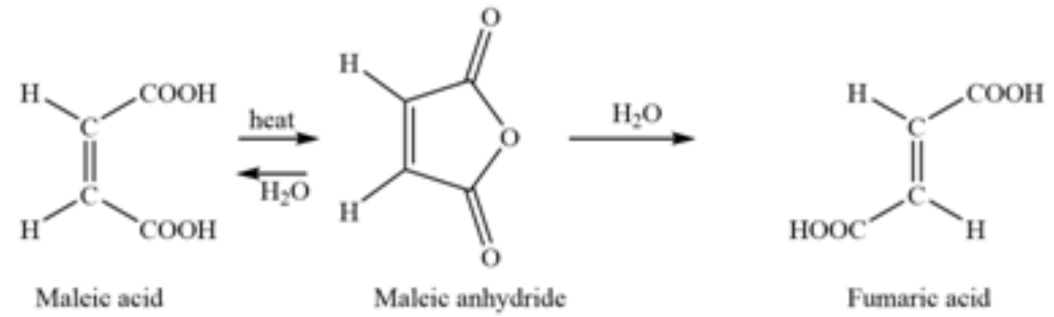
Cis-trans isomerism

Catalytic conversion of cis-trans isomers

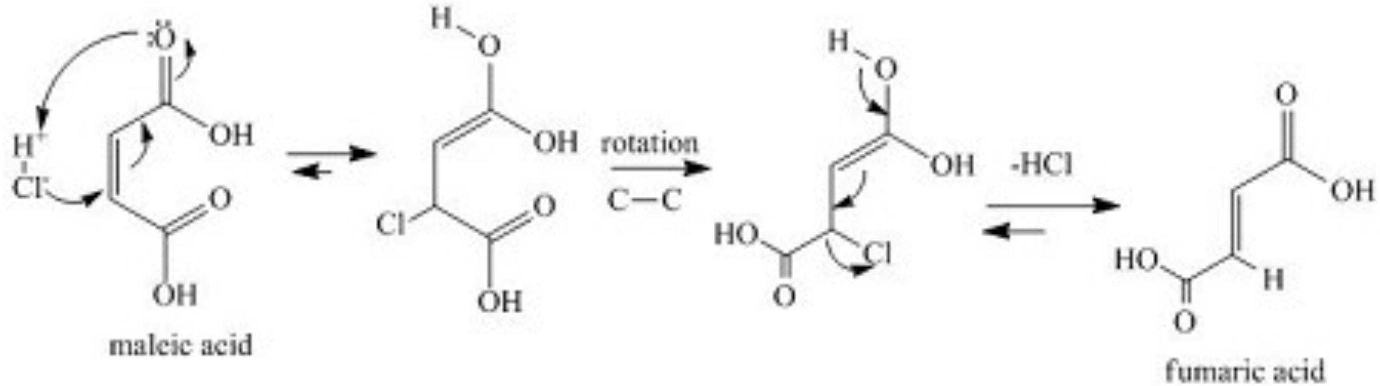


Mechanism of acid catalyzed conversion of cis-trans isomers

Heat



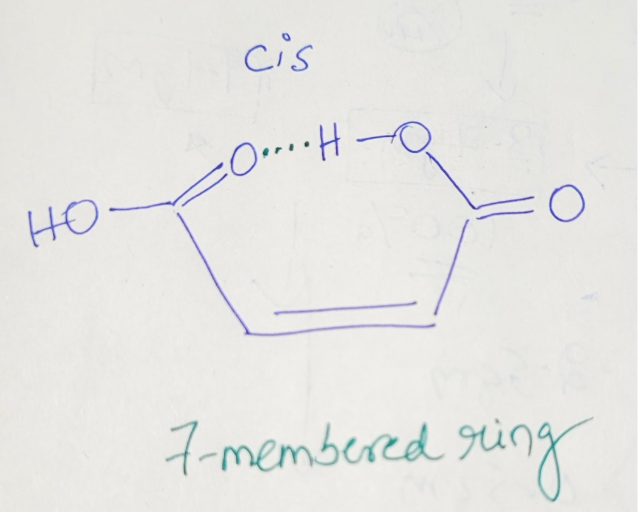
Acid catalyzed

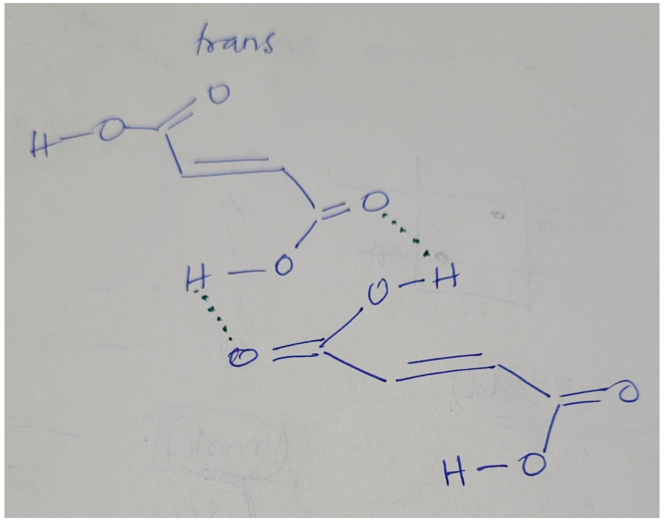


Different physical properties of cis-trans isomers

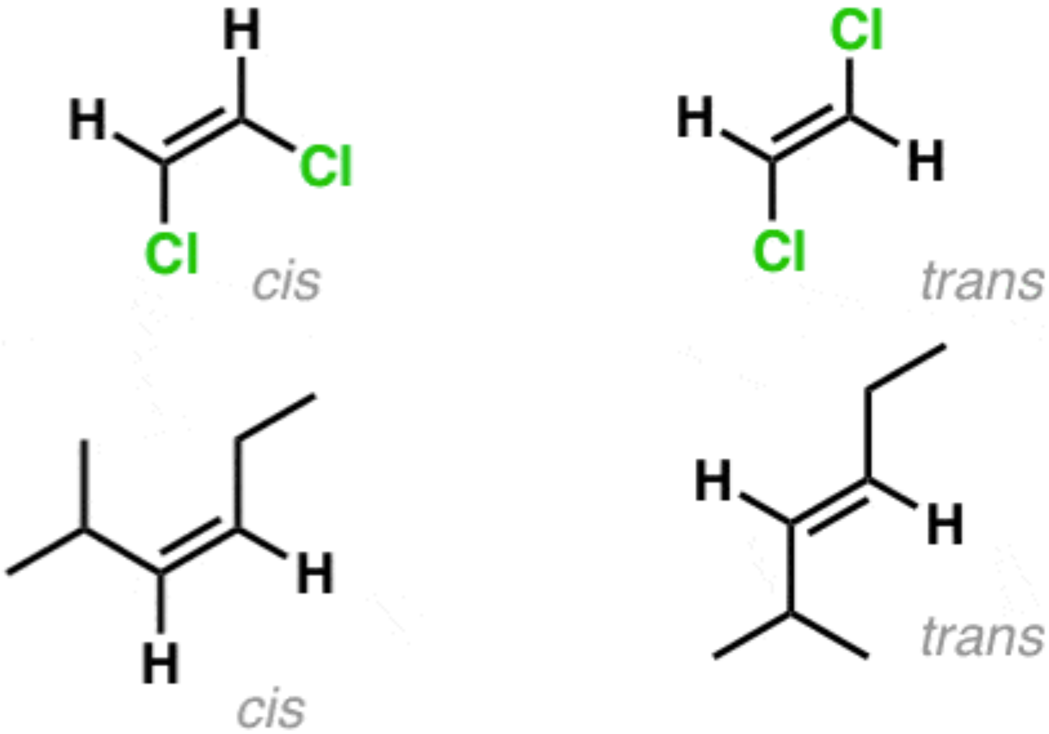
Fumaric acid (~290ºC) has a higher melting point than maleic acid (~130ºC). Explain why?

Hint: Inter vs intramolecular H-bonding
Draw the H-bonded structures

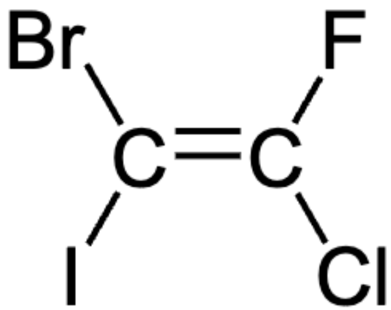


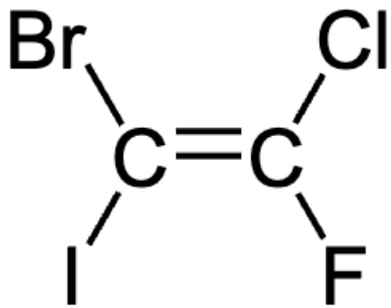


Nomenclature



Same functional groups





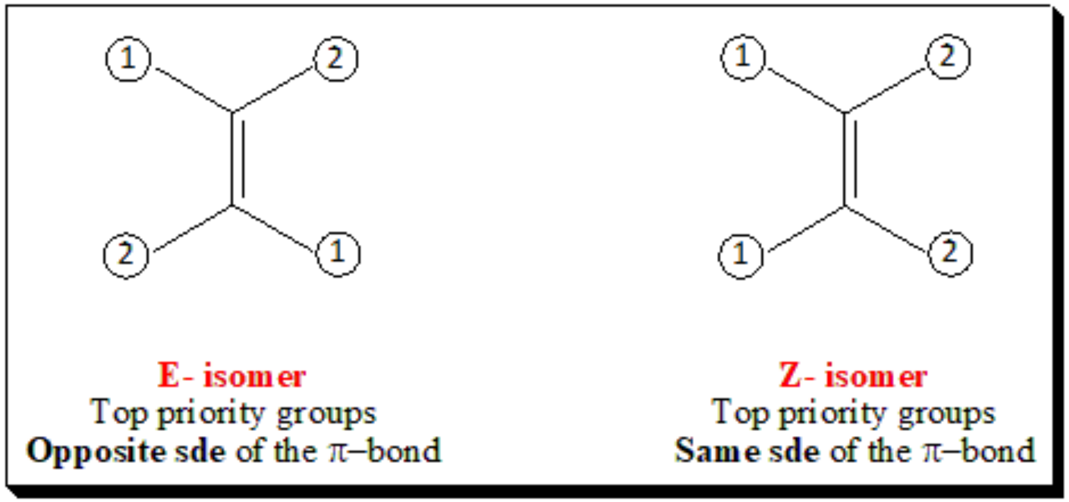
Cannot say cis or trans in this case?

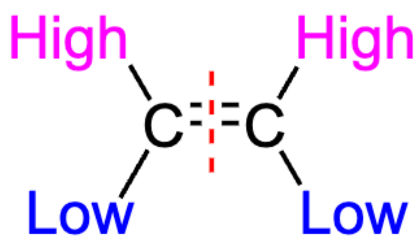
E, Z isomers

Assign priority to groups attached to both the sp^2 carbon

If high priority groups are on the same side, then Z (German *zusammen* = together)

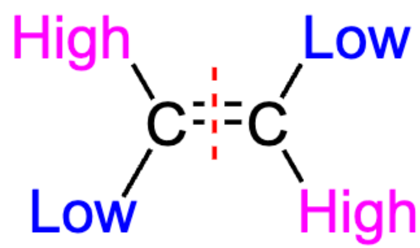
If high priority groups are on the opposite side, then E (German *entgegen* = opposite)





Z Configuration

High priority substituents are on the same side of the double bond



E Configuration

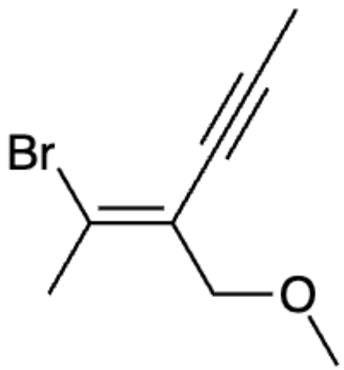
High priority substituents are on opposide sides of the double bond

Cahn-Ingold-Prelog (CIP) rule: Assigning priority to functional groups

Rule 1: The "first point of difference" rule

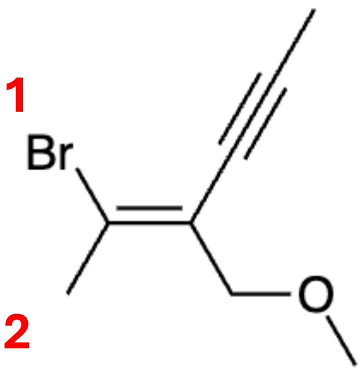
First, determine the two substituents on each double-bond carbon separately. Rank these substituents based on **the atom** which directly attached to the double-bond carbon. The substituent whose atom has a higher atomic number takes precedence over the substituent whose atom has a lower atomic number.

Which is higher priority, by the CIP rules: a C with an O and 2 H attached to it or a C with three C? The first C has one atom of high priority but also two atoms of low priority. How do these "balance out"? Answering this requires a clear understanding of how the ranking is done. The simple answer is that the first point of difference is what matters; the O wins.



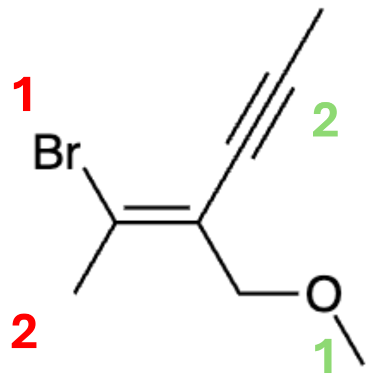
1

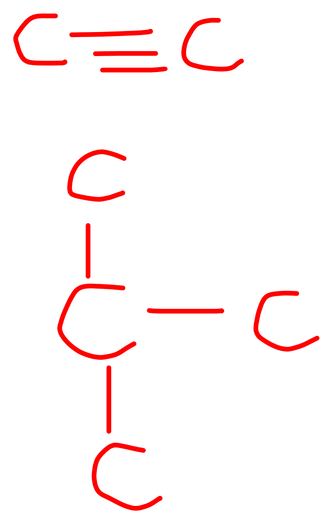
2



At the right hand end, the first atom attached to the double bond is a C at each position. A tie, so we look at what is attached to this first C. For the upper C, it is CCC (since the triple bond counts three times). For the lower C, it is OHH -- listed in order from high priority atom to low. OHH is higher priority than CCC, because of the first atom in the list. That is, the O of the lower group beats the C of the upper group. In other words, the O is the highest priority atom of any in this comparison; thus the O "wins".

Remember that atoms involved in multiple bonds are considered with a specific set of rules. These atoms are treated as if they have the same number of single-bond atoms as they have attached to multiply bonded atoms.

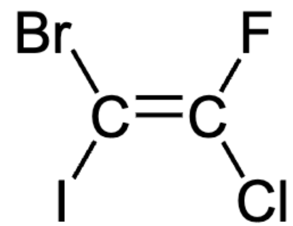


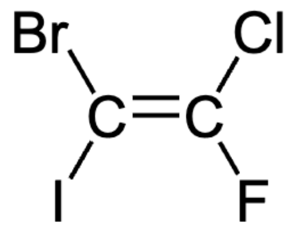


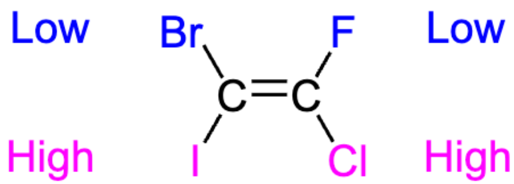
(E)-2-Bromo-3-(methoxymethyl)hex-2-en-4-yne

Rule 2

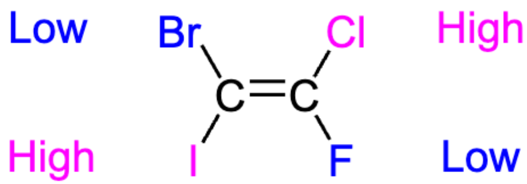
If the first atom on both substituents are the identical, then proceed along both substituent chains until the first point of difference is determined.







Z Configuration



E Configuration

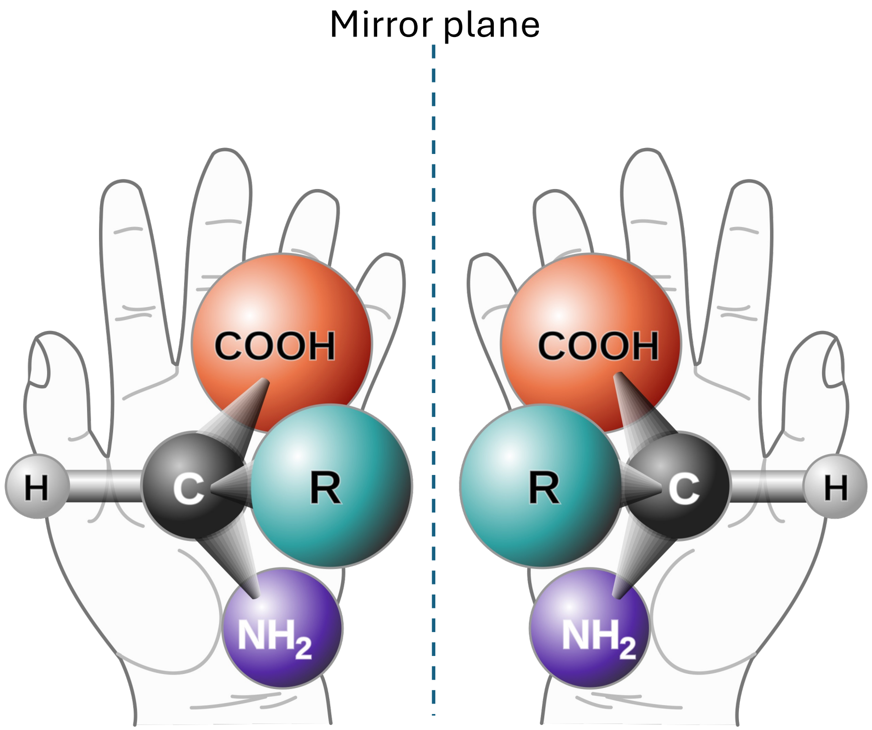
Types of stereoisomeism

Optical Isomerism

- Enantiomers
- Diastereomers

Chiral molecules - Enantiomers

Chirality is derived from the Greek word χερί (cherí), which means "hand"



Carbon atom connected to four different chemical groups form a chiral (stereo) center

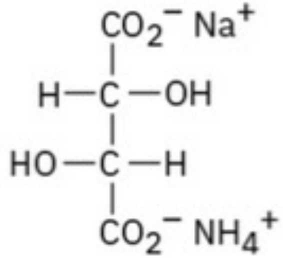
Do enantiomers differ in physical properties?

Do enantiomers differ in chemical properties?

How do we even know that enantiomers exist?

1848

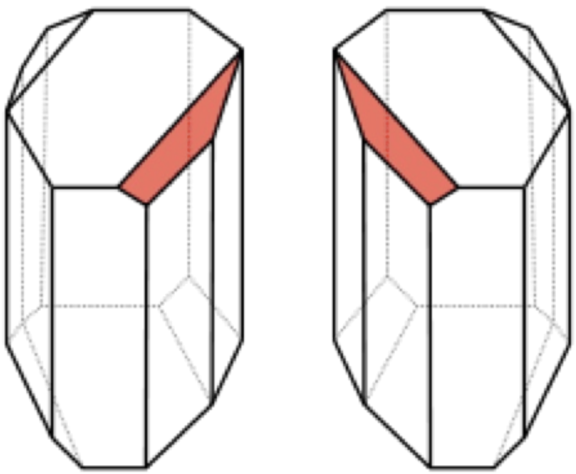
Crystallization of tartaric acid

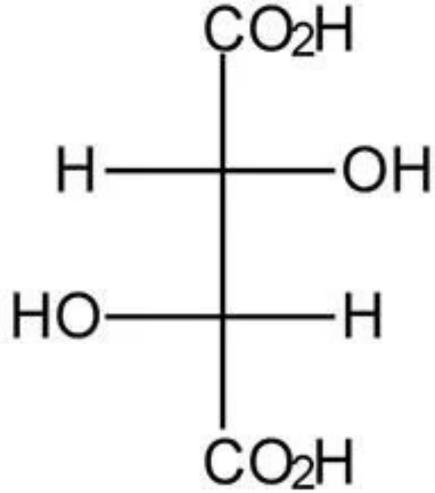


Sodium ammonium tartrate

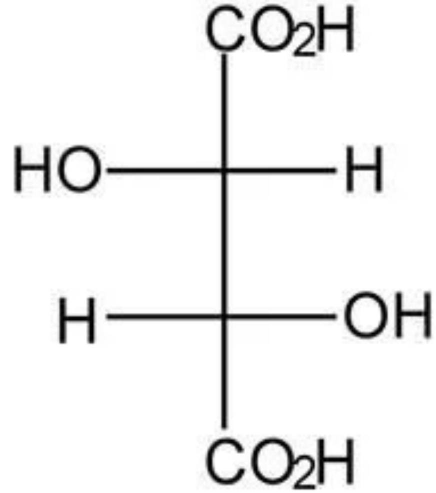
white

Two distinct crystals of tartaric acid





L(+) tartaric acid



D(-) tartaric acid

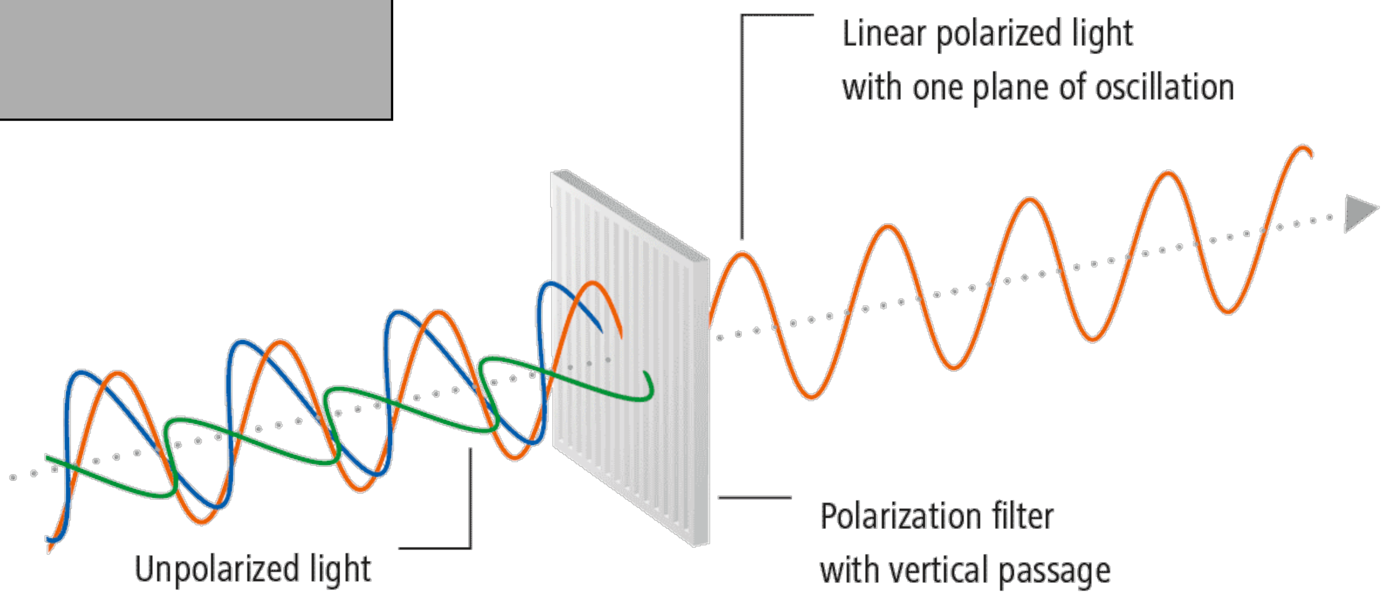
Enantiomers differ in their optical properties

Enantiomers also exhibit different biological properties

Polarized light

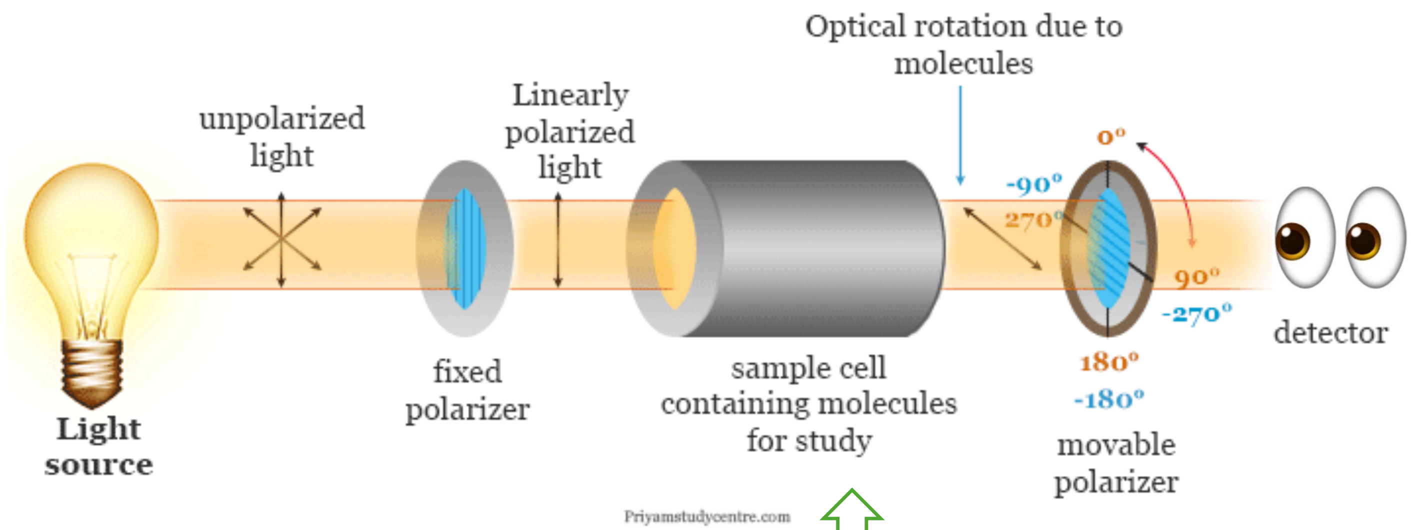
A light wave that is vibrating in more than one plane is known as unpolarized light.
Polarized light is made up of waves that vibrate in a single plane.

Three types of polarized light
Linear
Circular
Elliptical



In linearly polarized light, the electric vector of light moves in a single plane along the direction of propagation

Working principle of a Polarimeter



enantiomers

Based on optical properties

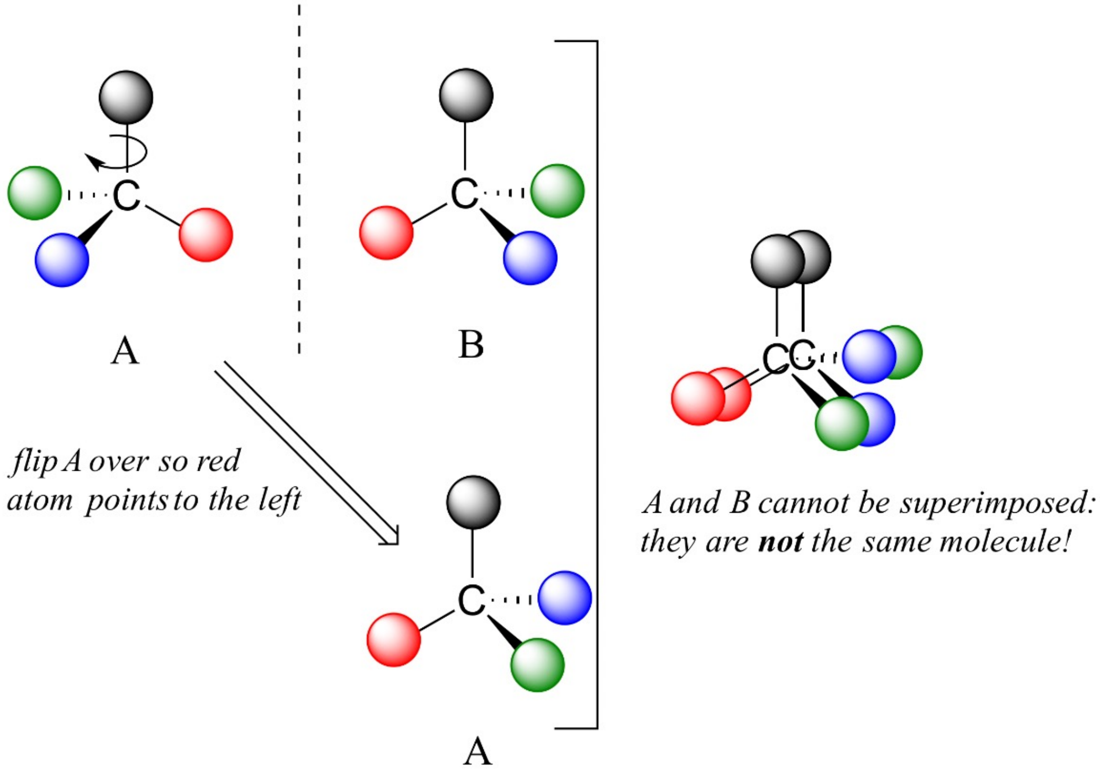
Chiral molecules: Not superimposable on its mirror image
(rotates PP light, does not have plane of symmetry)

Achiral molecules (Meso): Superimposable on its mirror image
(doesn’t rotate PP light, have a plane of symmetry)

Non-superimposable mirror images

Simple example is our hand

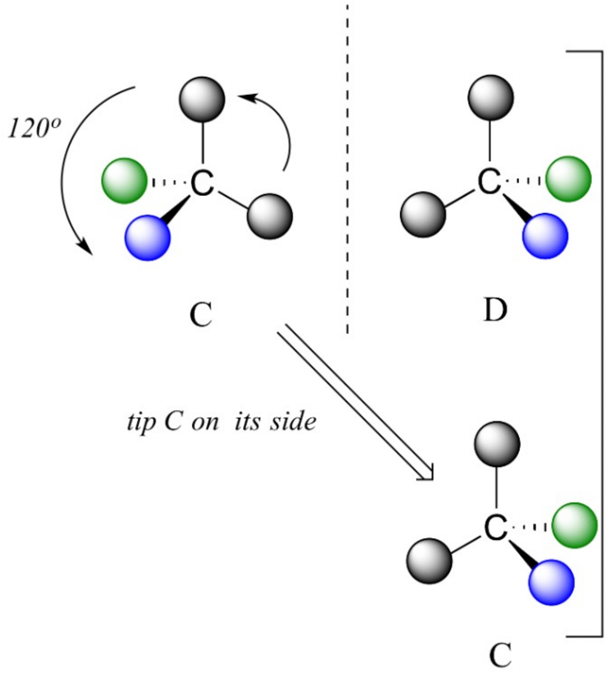
sp^3 hybridized C



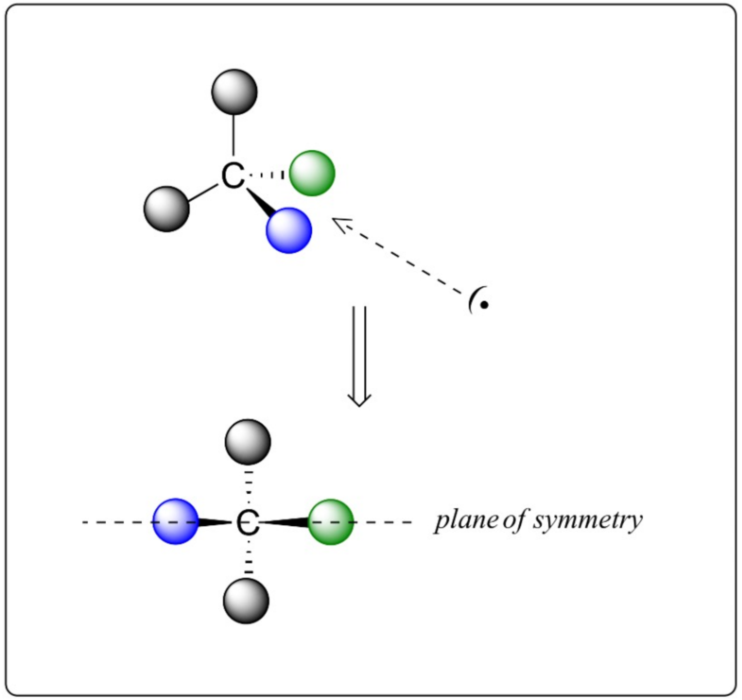
Carbon atom connected to four different chemical groups form a chiral (stereo) center

Superimposable mirror images

sp^3 hybridized C



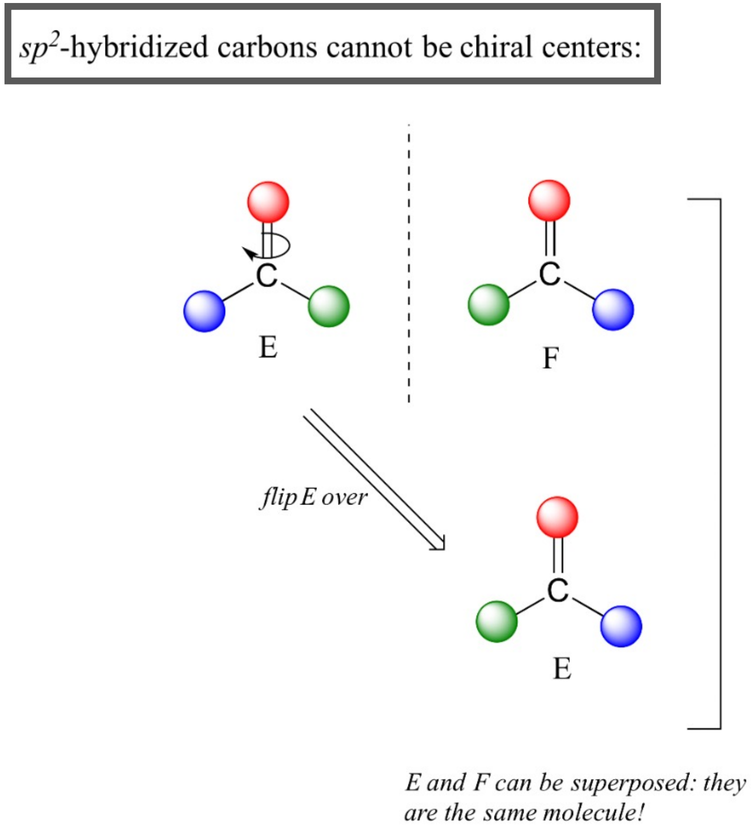
C and D can be superposed: they are the same molecule!

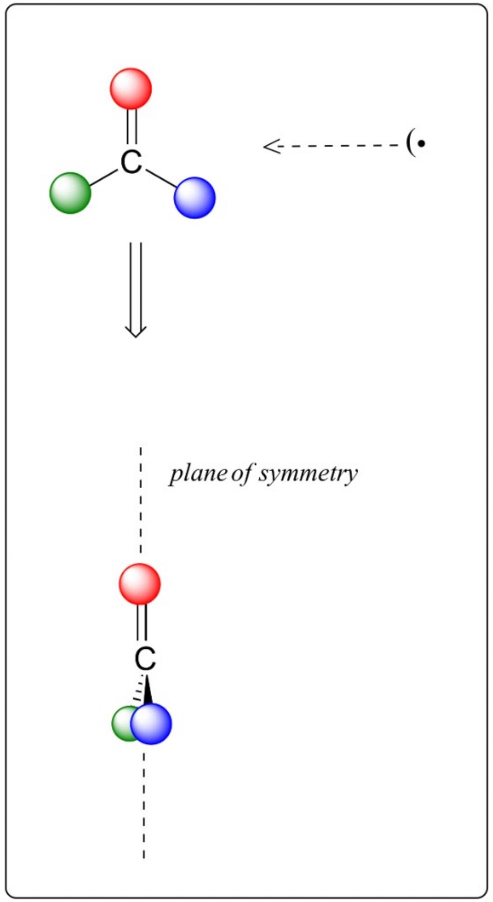


Carbon atom connected to at least two similar chemical groups

Here they are the grey spheres

Superimposable mirror images





Formaldehyde – achiral
Acetaldehyde – achiral

Chiral or achiral?

Symmetry elements (Point, Line and Plane)

Symmetry operations: A rotation or reflection or both – indistinguishable from original molecule

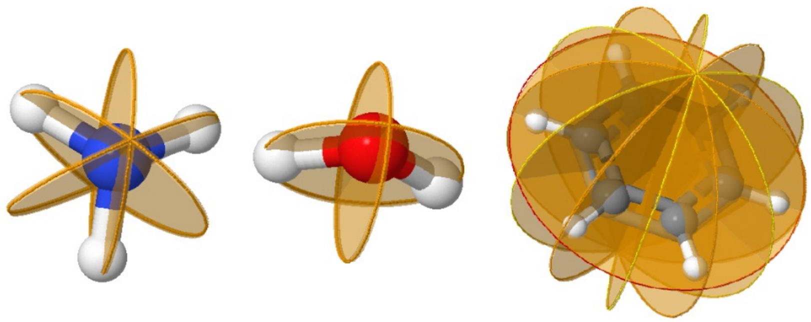
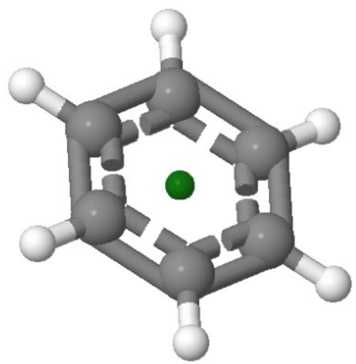
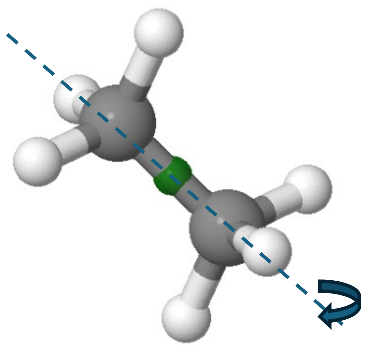
To be isomers, molecules must not be identical. **The test for "identicality" is one of superimposability**.

Rotational axis of symmetry (C_n): A rotation through an angle of 360°/n, which yields an image indistinguishable from the original.

Center of symmetry or inversion (**i**): Inversion through the center of symmetry leaves the molecule unchanged

Plane of symmetry ($\boldsymbol{\sigma}_p$): Reflection in the plane leaves the molecule unchanged

When these symmetry elements are present, the molecule is **Achiral**

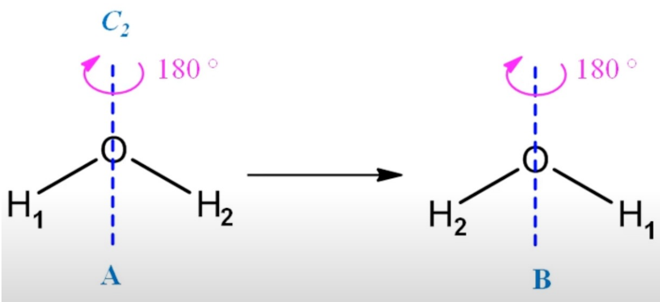
Benzene is a great example of a molecule having all the symmetry elements

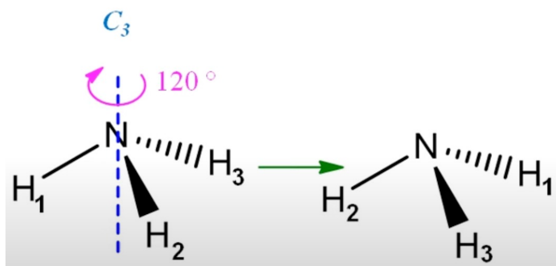
Proper axis/Rotational axis of symmetry (C_n)

An imaginary line passing through the center of mass around which a rotation by 360º/n leaves an object in an orientation indistinguishable from the original

C_2	$360°/2$	$= 180°$
C_3		$= 120°$
C_4		$= 90°$
C_5		$= 72°$
C_6		$= 60°$
$C_1 = E$		$= 360°$

Principal axis = C_n (n=highest) and passing through the maximum number of atoms





How can I dissolve the 2-fold axis of rotation in water keeping the molecule as water? (Hint: Isotope)

DIY

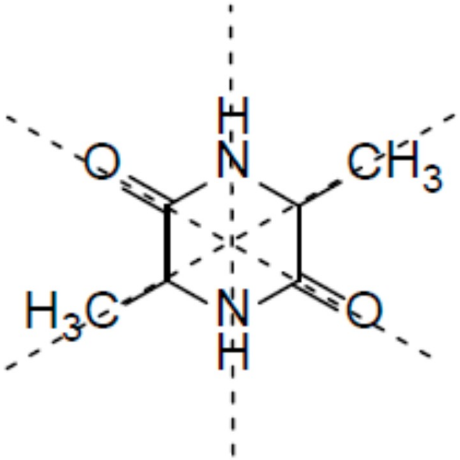
BF_3 molecule – how many C_n are there?
Which one is the principal axis?

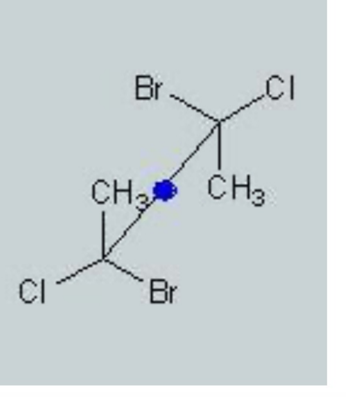
Methane molecule – how many C_n are there?
Which one is the principal axis?

Center of symmetry (i)

The center of symmetry (i), also called center of inversion, is a point in space such that if a line is drawn from any part (atom) of the molecule to that point and extended an equal distance beyond it, an analogous part (atom) will be encountered.

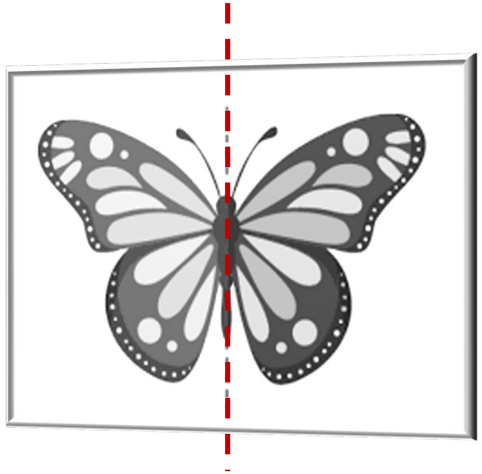
Example: The below molecule 3,6-dimethylpiperazine-2,5-dione (on the left) has center of symmetry located at the center of the molecule. Similarly, the molecule to the right has the center of symmetry located at the center of the C-C bond (marked by blue dot)

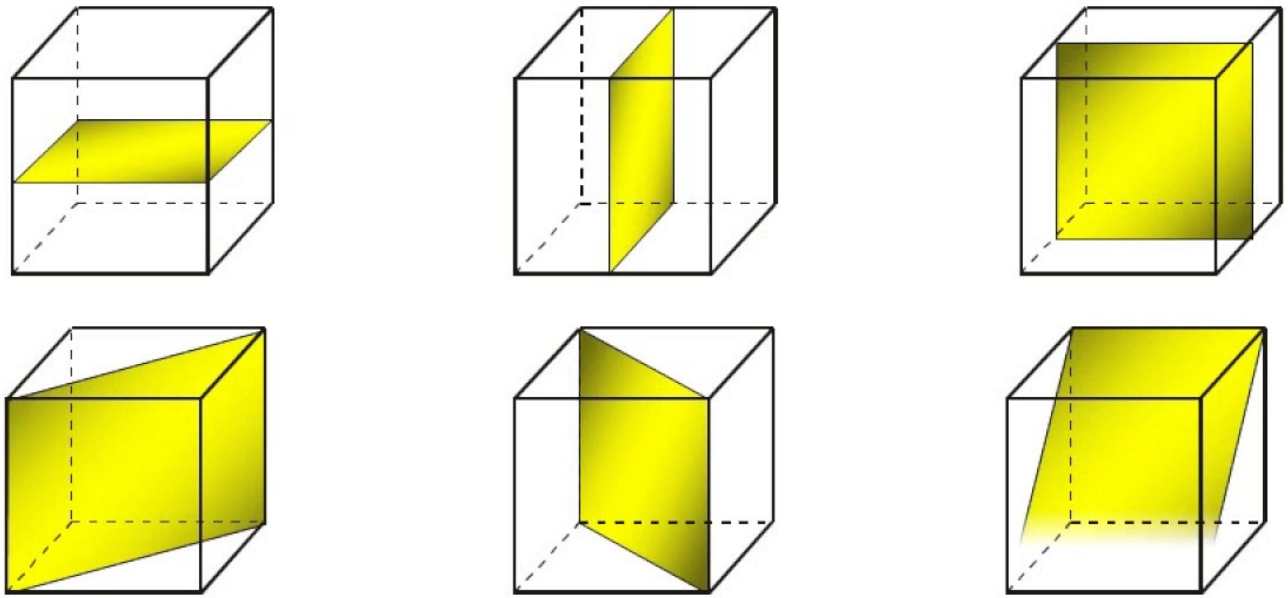




Center of symmetry can be located on an atom, a bond or a space in the molecule

Plane of symmetry





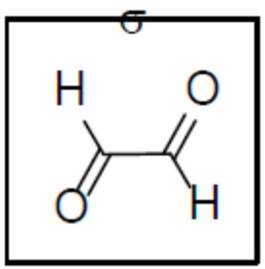
Plane of symmetry (σ)

A plane of symmetry (**σ**) is a reflection plane which brings into coincidence one point of the molecule with another one through the mirror reflection.

In other words, POS is an imaginary plane that bisects a molecule in such a way that the two halves of the molecule are mirror images of each other

Example: both trans and cis glyoxal have plane of symmetry

trans-glyoxal *cis*-glyoxal



In trans isomer – 1 plane of symmetry
In cis isomer – 2 planes of symmetry

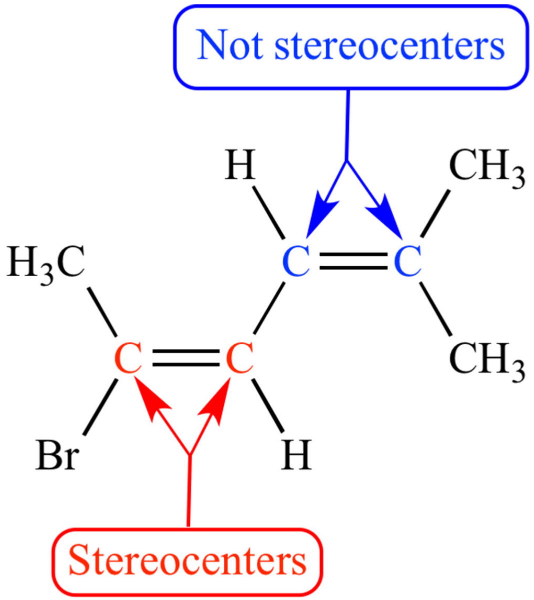
Stereocenters and Chiral centers

Stereocenters

A point in a molecule where interchanging any two groups creates a stereoisomer. Stereocenters can be atoms or groups of atoms, and can contain single or double bonds. They can exist in chiral or achiral molecules.
Might be a point in a molecule, not necessarily an atom

Chiral centers

A type of stereocenter where a carbon atom is bonded to four different groups, creating a non-superimposable mirror image. Chiral centers form enantiomers, which are molecules that are mirror images of each other.
A carbon center



sp^2 carbon cannot be chiral, but can be stereocenter

Test for chirality

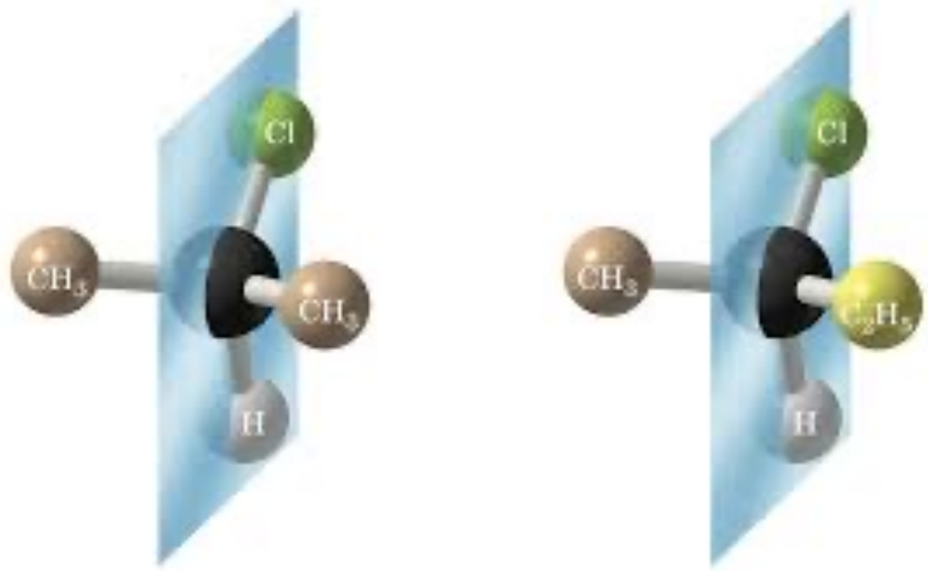
A molecule with either a plane of symmetry or a center of inversion cannot be chiral

A molecule with rotational symmetry can be chiral (e.g. allenes)

Basic test: Look for a chiral or a stereogenic center. In most of the cases, molecule which lacks a chiral center is achiral (though there are exceptions which you can find in the later slides)

If any two groups attached to a carbon atom are identical, the carbon atom is achiral

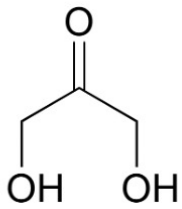
Plane of symmetry and chirality



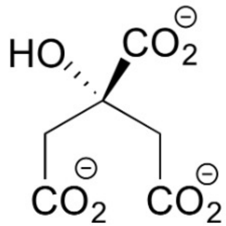
2-chloropropane vs 2-chlorobutane
Left side molecule has a POS, but not the right one. Symmetry is destroyed because carbon is attached to four different groups

Chiral carbon – sp^3 carbon attached to four different groups

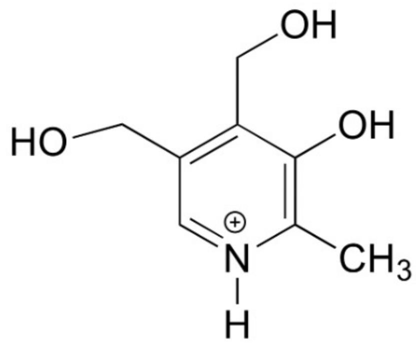
Biomolecules



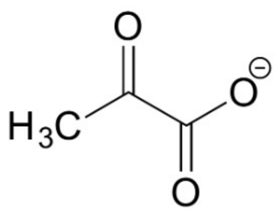
dihydroxyacetone



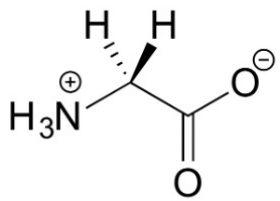
citrate



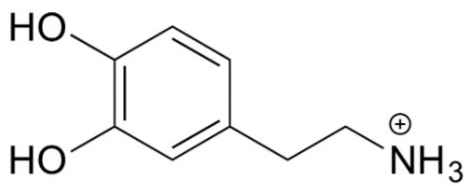
pyridoxine
(vitamin B_6)



pyruvate



glycine
(amino acid)



dopamine
(neurotransmitter)

Convince yourself that these molecules are all ACHIRAL